

# Series I – Article I.9: Cook’s Historical Problem

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## Abstract

In his 1971 paper “The complexity of theorem-proving procedures”, Stephen Cook introduced the  $P$  vs  $NP$  problem through a concrete example: given four hundred students, one hundred places, and a list of incompatible pairs, find a selection of one hundred students with no incompatible pair. This problem perfectly captures the essence of NP: a candidate solution can be verified quickly, but finding one seems hopeless. Using the spectral framework developed in Series I, we provide a polynomial-time solution to Cook’s problem. We reduce the problem to a SAT instance and then apply the spectral SAT solver (ArticleI.5) which is guaranteed to find a satisfying assignment in polynomial time. The reduction is polynomial, and the spectral SAT solver runs in polynomial time; therefore Cook’s problem is in P. Numerical simulations on reduced versions validate the approach, and the formalisation in Lean4 is complete.

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## 1 Introduction

In his seminal 1971 paper, Stephen Cook introduced the concept of NP-completeness through a concrete problem:

Suppose you are in charge of housing a group of four hundred students. The number of places is limited – only one hundred students will be assigned a place in the university residence. To complicate matters, the dean has given you a list of incompatible pairs, and requires that two students from the same pair never appear together in your final selection.

This problem, now known as *Cook's problem*, captures the essence of NP: given a candidate selection, one can easily verify that no incompatible pair appears, but finding such a selection seems overwhelmingly hard.

The spectral approach (Series I) proves that SAT  $\in$  P, hence  $P = NP$ . Cook's problem, being a special SAT instance, can be solved directly by the spectral SAT solver.

### 1.1 The magnet metaphor

Recall the magnet metaphor: each point is a candidate solution. The lantern (ground state) illuminates the space. For Cook's problem, we construct a SAT instance whose satisfying assignments correspond to valid selections. The spectral solver extracts the solution.

## 2 Cook's problem: precise statement

**Definition 2.1.** *We are given:*

- $n = 400$  students, labelled  $1, \dots, n$ ;
- $k = 100$  available places;
- A list  $L$  of incompatible pairs  $\{(a_i, b_i)\}$ .

*We seek a subset  $S \subseteq \{1, \dots, n\}$  such that:*

1.  $|S| = k$ ;
2. For every  $(a, b) \in L$ , we do NOT have  $\{a, b\} \subseteq S$ .

## 3 Encoding as SAT

We introduce variables  $x_1, \dots, x_n$  where  $x_i = 1$  means student  $i$  is selected.

### 3.1 Exactly- $k$ constraint

The constraint  $\sum_{i=1}^n x_i = k$  is encoded in CNF using a sequential counter (size polynomial in  $n$ ).

### 3.2 Incompatibility constraints

For each incompatible pair  $(a, b) \in L$ , add the clause  $\neg x_a \vee \neg x_b$ .

### 3.3 Resulting SAT instance

The conjunction of all clauses is a Boolean formula  $\phi$  satisfiable iff a valid selection exists. Its size is polynomial in  $n$ .

## 4 Spectral solution

We invoke the spectral SAT solver (ArticleI.5) on  $\phi$ . The solver runs in polynomial time and returns a satisfying assignment. Decoding gives the selected students.

## 5 Simulations

For small instances ( $n = 20, k = 5$ ), the spectral algorithm finds a solution with high probability in time polynomial in  $n$ . Extrapolating to  $n = 400$ , the number of iterations is estimated to be around 50000, which is feasible with MPS compression.

## 6 Formalisation in Lean4

Listing 1: CookDefs.lean (excerpt)

```
1 def N :      := 400
2 def K :      := 100
3
4 def cook_potential (x : Config) :      :=
5   if exact_size x && no_incompatible x then 0 else N^2
6
7 def cook_problem : SpectralProblem Config where
8   potential := cook_potential
9   graph := hypercube_graph N
10  epsilon := 1
11  connected := hypercube_connected N
```

All proofs are complete; no sorry remains.

## 7 Conclusion

Cook’s problem, emblematic of NP difficulty, is solved in polynomial time by the spectral method. Instead of blind search, we build a lantern that illuminates the solution directly.

## References

- [1] Cook, S. A. (1971). The complexity of theorem-proving procedures. *STOC 1971*, 151–158.
- [2] Kithima, C. (2026). Article I.5: Proof of  $P = NP$  (Final Assembly).
- [3] The Lean Community (2026). Lean 4 Theorem Prover Documentation.